

Thermal Cycling Validation Test Specification

Project: Project Falcon

Product: PressureSense X1

Status: RELEASED DEMO SPECIFICATION

Primary method: ENV-TC-004

1. Purpose and scope

This synthetic demonstration specification defines the validation intent, sample population, test sequence, operating conditions and acceptance criteria for PressureSense X1. It is provided only as seeded LabOS prototype content.

2. Test definition

Parameter	Requirement
Primary condition	-40 C to +150 C
Exposure / sequence	1000 cycles
Sample population	10 DUTs minimum; genealogy maintained by LabOS
Preconditioning	24 h at laboratory ambient unless otherwise specified
Data capture	Record test conditions, equipment, calibration, operator, DUT and evidence
Acceptance	8.0 to 14.0 V functional margin

3. Readiness requirements

Before execution: confirm the released method revision, correct DUT population, valid calibrated equipment, qualified operator, predecessor completion, sample availability and approved acceptance criteria. Any deviation shall be recorded as a LabOS issue and shall trigger replanning when it affects availability or duration.

4. Evidence and reporting

Raw data, result parameters, pass/fail evaluation, photographs where relevant, equipment and calibration traceability, operator identity and final report evidence shall be linked to the applicable test leg and validation requirements.

DEMO DOCUMENT - not a released engineering specification for production use.